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ELECTRIC VEHICLE WITH A COOLING ARRANGEMENT

Abstract

An electric vehicle including a frame; at least one ground-engaging member operatively connected to the frame; an electric motor operatively connected to the at least one ground-engaging member; an electric powerpack supported by the frame, the powerpack including: a battery pack defining a battery cooling channel; and a plurality of battery cells housed in the battery housing; a charger defining a charger cooling channel; an inverter including an inverter cooling channel, the battery cooling channel, the charger cooling channel, and the inverter cooling channel being fluidly connected together by rigid fluid connections; radiators fluidly connected to the battery cooling channel; a cooling circuit being formed at least in part by the battery cooling channel, the charger cooling channel, the inverter cooling channel, and the at least one radiator; and a pump fluidly connected to the cooling circuit for pumping liquid coolant through the cooling circuit.

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Background/Summary

CROSS-REFERENCE [0001] The present application claims priority from U.S. Provisional Patent Application No. 63/273,435, entitled “Electric Vehicle with a Cooling Arrangement,” filed Oct. 29, 2021, and to U.S. Provisional Patent Application No. 63/273,468, entitled “Electric Vehicle with an Electric Powerpack Arrangement,” filed Oct. 29, 2021, the entirety of both of which is incorporated by reference herein.

FIELD OF TECHNOLOGY
[0002] The present technology relates to electric vehicles having cooling arrangements, specifically straddle-seat electric vehicles.
BACKGROUND
[0003] Straddle seat vehicles, including motorcycles, all-terrain vehicles, and snowmobiles, are popular transport and recreational vehicles. As the move toward electrification of vehicles progresses, interest in electric versions of straddle seat vehicles increases. Electric vehicles generally include components that are required to be maintained within a specified temperature range, for instance battery packs, inverters, and motors. In some vehicles, cooling is provided through air-flow based cooling arrangements.
[0004] To better address cooling of each component, some solutions include circulating a liquid coolant around a circuit in thermal communication with the heat-generating components and one or more heat exchangers. In order to circulate the liquid coolant through these heat generating components, as well as to and from the heat exchanger(s), hoses are often employed.
[0005] Applied to straddle seat electric vehicles, however, hoses manifest several disadvantages. For example, hoses generally have a minimum radius of curvature which limits how much the hose can be bent. In the compact powerpack arrangements necessary for straddle seat vehicles, curvature limitations can severely restrict the possible layout design of the different powerpack components. Inclusion of hoses extending between different components further occupies the limited space. It is further noted that each hose and hose connection adds complication in terms of fabrication and maintenance. Each hose and hose connection could be a point of failure (in the form of material failure like cracking or leaks) and requires installation for each liquid connection via clamp, collar or other mechanical means.
[0006] There is therefore a desire for cooling arrangements for electric straddle seat vehicles addressing at least some of the above described disadvantages.
SUMMARY
[0007] It is an object of the present technology to ameliorate at least some of the inconveniences

present in the prior art.

[0008] According to one aspect of the present technology, there is provided an electric vehicle, specifically an electric motorcycle and an electric snowmobile, with an electric powerpack and a liquid cooling circuit. Portions of the cooling circuit within the powerpack, specifically cooling channels of the battery pack, the charger, and the inverter, are fluidly connected together through rigid connections, including by portions of the battery pack housing, the charger housing, and the inverter housing. In this way, no hoses or tubing is required within the powerpack, thereby reducing the number of parts to be assembled and reducing the number of points of possible leaks within the powerpack. A coolant reservoir is further rigidly fluidly connected to the cooling channels of the battery pack, the charger, and the inverter.

[0009] According to one aspect of the present technology, there is provided an electric vehicle including a frame; at least one ground-engaging member operatively connected to the frame; an electric motor operatively connected to the at least one ground-engaging member; an electric powerpack supported by the frame, the powerpack including: a battery pack including: a battery housing defining a battery cooling channel; and a plurality of battery cells housed in the battery housing; a charger electrically connected to the plurality of battery cells, the charger including: a charger housing defining a charger cooling channel; an inverter electrically connected to the plurality of battery cells, the inverter including: an inverter housing defining an inverter cooling channel, the battery cooling channel, the charger cooling channel, and the inverter cooling channel being fluidly connected together by rigid fluid connections; at least one radiator fluidly connected to the battery cooling channel; a cooling circuit being formed at least in part by the battery cooling channel, the charger cooling channel, the inverter cooling channel, and the at least one radiator; and a pump fluidly connected to the cooling circuit for pumping liquid coolant through the cooling circuit.

[0010] In some embodiments, the vehicle further includes a motor cooling channel for cooling the electric motor fluidly connected to the battery cooling channel and forming a portion of the cooling circuit.

[0011] In some embodiments, the vehicle further includes a swing arm pivotally connected to the frame, the swing arm including a swing arm housing; and the electric motor includes a motor housing; the motor housing includes a channeled outer surface in thermal contact with at least some internal components of the electric motor; the swing arm housing defines a motor cavity; the electric motor being disposed in the motor cavity; and the motor cooling channel is formed between the channeled outer surface of the motor housing and an internal surface of the motor cavity.

[0012] In some embodiments, the vehicle further includes at least one flexible tube fluidly connecting the motor cooling channel to the pump.

[0013] In some embodiments, a channel formed by the channeled outer surface forms a spiral winding around an exterior of the electric motor.

[0014] In some embodiments, the inverter housing is fastened to the battery housing; and at least one of an inlet and an outlet of the inverter cooling channel is connected to at least one of an inlet and an outlet of the battery cooling channel.

[0015] In some embodiments, the battery pack includes a first electric connector electrically connected to the plurality of battery cells and disposed on an exterior of the battery housing; the inverter includes a second electric connector disposed on an exterior of the invertor housing; the first electric connector and the second electric connector are selectively connected together; and when the vehicle is in operation, the inverter receives electric power from the plurality of battery cells via the first electric connector and the second electric connector.

[0016] In some embodiments, the second electric connector and the at least one of the inlet and the outlet of the inverter cooling channel are disposed on a same side of the inverter.

[0017] In some embodiments, the charger and the inverter are mounted to the battery housing.

[0018] In some embodiments, the vehicle further includes a coolant reservoir connected to the powerpack.
[0019] In some embodiments, the vehicle further includes at least one of the rigid fluid connections connecting the battery cooling channel, the charger cooling channel, and the inverter cooling channel together is formed by one of the charger housing, the battery housing, and the inverter housing.
[0020] In some embodiments, connections between the battery cooling channel, the charger cooling channel, and the inverter cooling channel are internal to the electric powerpack.
[0021] In some embodiments, the charger cooling channel extends along a side of the charger facing an interior of the powerpack; and the inverter cooling channel extends along a side of the inverter facing the interior of the powerpack.
[0022] In some embodiments, the battery cooling channel extends through a center portion of the battery pack in the interior of the powerpack.
[0023] In some embodiments, the at least one radiator includes: a left radiator disposed on a left side of the vehicle; and a right radiator disposed on a right side of the vehicle.
[0024] In some embodiments, the vehicle further includes a plurality of flexible tubing components fluidly connecting the left radiator and the right radiator to the at least one battery cooling channel and forming a portion of the cooling circuit.
[0025] In some embodiments, at least one of the left radiator and the right radiator includes a fan connected to a radiator housing thereof.
[0026] In some embodiments, the vehicle further includes a coolant reservoir fluidly connected to the cooling circuit.
[0027] In some embodiments, the charger housing includes at least one of an inlet and an outlet configured to be sealingly fit into at least one of an outlet and an inlet of the coolant reservoir.
[0028] In some embodiments, the coolant reservoir is disposed at least partially forward of the electric powerpack; and the pump is disposed at least partially rearward of the electric powerpack.
[0029] In some embodiments, the vehicle further includes a motor cooling channel for cooling the electric motor fluidly connected to the battery cooling channel, and a coolant reservoir connected to the powerpack; and the at least one radiator includes a pair of radiators, and the cooling circuit is further formed by at least the pair of radiators, the motor cooling channel, and the coolant reservoir.
[0030] In some embodiments, the vehicle further includes a straddle seat; and the at least one ground-engaging member includes: at least one front ground-engaging member disposed at least in part forward of the electric powerpack, and at least one rear ground-engaging member disposed at least in part rearward of the electric powerpack.
[0031] In some embodiments, the at least one ground-engaging member includes: a front wheel disposed at least in part forward of the electric powerpack, and a rear wheel disposed at least in part rearward of the electric powerpack; and the vehicle is an electric motorcycle.
[0032] In some embodiments, the vehicle is an electric snowmobile; and the at least one ground-engaging member includes two skis.
[0033] According to another aspect of the present technology, there is provided an electric vehicle having: a frame, the frame having a pair of frame members; a front ground-engaging member operatively connected to the frame; a swing arm pivotally connected to the pair of frame members about a swing arm pivot axis, the swing arm pivot axis extending through the pair of frame members; a rear ground-engaging member operatively connected to the swing arm; an electric motor mounted to the swing arm and operatively connected to the rear ground-engaging member; a motor cooling channel in thermal communication with the electric motor for cooling the electric motor; a battery pack supported by the frame; a battery cooling channel in thermal communication with the battery pack for cooling the battery pack; and at least one hose fluidly connected between the battery cooling channel and the motor cooling channel, the at least one hose extending in a space defined laterally between the frame members and longitudinally between the battery pack

and the motor.
[0034] In some embodiments, the swing arm pivot axis extends through the space defined laterally between the frame members and longitudinally between the battery pack and the motor.
[0035] In some embodiments, with the vehicle at rest: a front end of the at least one hose is forward of the swing arm pivot axis; the front end of the at least one hose is vertically higher than the swing arm pivot axis; a rear end of the at least one hose is rearward of the swing arm pivot axis; and the rear end of the at least one hose is at least in part vertically lower than the swing arm pivot axis.
[0036] In some embodiments, part of the at least one hose is disposed at a radial distance from the swing arm pivot axis that is less than twice a diameter of the at least one hose.
[0037] In some embodiments, the radial distance is less than 1.5 times the diameter of the at least one hose.
[0038] In some embodiments, the radial distance is less than the diameter of the at least one hose.
[0039] In some embodiments, the at least one hose includes a first hose and a second hose; one of the first and second hoses supplying liquid coolant from the battery cooling channel to the motor cooling channel; and another one of the first and second hoses supplying liquid coolant from the motor cooling channel to the battery cooling channel.
[0040] In some embodiments, the vehicle also has a pump fluidly connected between the first hose and the battery cooling channel.
[0041] In some embodiments, the battery pack includes: a battery housing defining the battery cooling channel; and a plurality of battery cells housed in the battery housing.
[0042] In some embodiments, the battery pack is completely forward of the swing arm pivot axis.
[0043] For the purposes of the present application, terms related to spatial orientation such as forward, rearward, front, rear, upper, lower, left, and right, are as they would normally be understood by a driver of the vehicle sitting therein in a normal driving position with the vehicle being upright and steered in a straight ahead direction.
[0044] Implementations of the present technology each have at least one of the above-mentioned object and/or aspects, but do not necessarily have all of them. It should be understood that some aspects of the present technology that have resulted from attempting to attain the above-mentioned object may not satisfy this object and/or may satisfy other objects not specifically recited herein.
[0045] Additional and/or alternative features, aspects and advantages of implementations of the present technology will become apparent from the following description, the accompanying drawings and the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0046] For a better understanding of the present technology, as well as other aspects and further features thereof, reference is made to the following description which is to be used in conjunction with the accompanying drawings, where:
[0047] FIG. **1** is a top, rear, right side perspective view of an electric motorcycle according to a non-limiting embodiment of the present technology;
[0048] FIG. **2** is a top plan view of the vehicle of FIG. **1**;
[0049] FIG. **3** is left side elevation view of the vehicle of FIG. **1**;
[0050] FIG. **4** is a right side elevation view of the vehicle of FIG. **1**, with body panels having been removed;
[0051] FIG. **5** is a left side elevation view of the vehicle of FIG. **1**, with body panels having been removed;
[0052] FIG. **6** is a top, front, left side perspective view of a frame, a powerpack, a drivetrain, and a swing arm of the vehicle of claim **1**, with a housing cover of the swing arm having been removed;

[0053] FIG. **7** is a top, rear, right side perspective view of the components of FIG. **6**;
[0054] FIG. **8** is a top, rear, right side, exploded, perspective view of a motor and the swing arm of the vehicle of FIG. **1**;
[0055] FIG. **9** is a top plan view of the components of FIG. **6** with the housing cover of the swing arm;
[0056] FIG. **10** is a left side elevation view of the components of FIG. **9**;
[0057] FIG. **11** is a schematic diagram of a cooling circuit of the vehicle of FIG. **1**;
[0058] FIG. **12** is a top, front, left side perspective view of the powerpack, the motor, and the cooling circuit components of the vehicle of FIG. **1**;
[0059] FIG. **13** is a cross-sectional view of portions of the powerpack of FIG. **6**, taken along line **13-13** of FIG. **9**;
[0060] FIG. **14** is a bottom, rear, left side perspective view of a charger of the powerpack of the vehicle of FIG. **1**;
[0061] FIG. **15** is a cross-sectional view of portions of the powerpack of FIG. **6**, taken along line **15-15** of FIG. **10**;
[0062] FIG. **16** is a cross-sectional view of portions of the powerpack of FIG. **6**, taken along line **16-16** of FIG. **10**, with battery cells therein being illustrated schematically;
[0063] FIG. **17** is a rear, left side perspective view of a battery pack of the powerpack of FIG. **6**;
[0064] FIG. **18** is a top, rear, right side perspective view of an inverter and a rigid connector of the powerpack of FIG. **6**; and
[0065] FIG. **19** is a right side elevation view of an electric snowmobile according to another non-limiting embodiment of the present technology.
[0066] It should be noted that, unless otherwise explicitly specified herein, the drawings are not necessarily to scale.

DETAILED DESCRIPTION

[0067] The present technology will be described herein with respect to a straddle-seat electric vehicles, specifically a two-wheeled electric motorcycle **100** and an electric snowmobile **300**. Aspects of the present technology could also be implemented in different straddle-seat electric vehicles, such as three-wheeled electric vehicles and all-terrain vehicles (ATVs).
[0068] While the motorcycle **100** illustrated herein is a trail style electric motorcycle **100**, it is contemplated that motorcycles according to the present technology could vary by a plurality of vehicle characteristics. These vehicle characteristics could include, but are not limited to, a rider posture configuration (also referred to as a rider position), a motorcycle type, tire type, a wheelbase, a steering arrangement, a weight distribution, a squat ratio, a rake angle, a seat height, and a mechanical trail. The rider posture configuration, or rider position, is the relative spacing and position of a rider's hands (when holding the handlebars), the rider's feet (when positioned on the footrests) and the rider's buttocks (when the rider is seated on a seat of the motorcycle). The steering arrangement could also vary and can be described by a variety of parameters, including but not limited to: a length of front suspension travel, a length of rear suspension travel, a front suspension stiffness, a rear suspension stiffness, a front and/or rear wheel size, rake angle, mechanical trail, triple clamp offset, squat ratio, and wheel base.
[0069] With reference to FIGS. **1** to **5**, the electric motorcycle **100**, referred to herein as the vehicle **100**, has a front end **102**, a rear end **104**, and a longitudinal centerplane **103** defined consistently with the forward travel direction of the vehicle **100**.
[0070] The vehicle **100** has a frame **110**, shown in additional detail in FIGS. **6** to **8**. The frame **110** includes a front suspension receiving portion **112**, specifically a tube **112**, sometimes referred to as a “head tube”, for receiving therethrough a front fork assembly **124** (described in more detail below). Extending rearward from the tube **112**, the frame **110** includes forward frame members **114**. In the illustrated embodiment, there are six forward frame members **114** (three on each side of the centerplane **103**) but it is contemplated that the specific number and arrangement of forward

frame members **114** could vary.

[0071] The frame **110** also includes two upper intermediate frame members **116** extending rearward from the forward members **114**. The frame members **116** are generally hockey-stick shaped, with rear portions of the members **116** curving rearward and downward from generally horizontal forward portions of the members **116**. In different embodiments, the frame members **116** could be differently shaped.

[0072] The frame **110** further includes two lower intermediate frame members **118** connected to rear ends of the frame members **116**. The frame members **118** extend generally vertically along left and right sides of the vehicle **100**. The frame members **118** are generally in the shape of flattened boomerangs, but the particular shape could vary. It is contemplated that the frame members **118** could be left and right sides of a common frame bracket.

[0073] The frame **110** further includes a rear frame structure **120** (FIGS. **4** and **5**) connected to and extending rearward and slightly upward from the upper intermediate frame members **116** and the lower intermediate frame members **118** (omitted from FIGS. **6** to **8**). The rear frame structure **120** is also referred to as a seat support structure **120**. It is contemplated that the frame **110** could include additional frame members, including but not limited to additional intermediate frame members and additional rear frame members.

[0074] The vehicle **100** is a two-wheeled vehicle **100** including a front wheel **121** and a rear wheel **127**. The front wheel **121** and the rear wheel **127** each have a tire secured thereto. The front wheel **121** and the rear wheel **127** are centered with respect to the longitudinal centerplane **103**.

[0075] The front wheel **121** is connected to the frame **110** by a front suspension assembly **123**. The front suspension assembly **123** includes a front fork assembly **124** for supporting the front end **102** of the vehicle **100**. The front fork assembly **124** includes a triple clamp assembly **125** connected to the tube **112** of the frame **110**. The front fork assembly **124** includes a pair of front shocks **122** connected to the triple clamp assembly **125**. The front wheel **121** of the front fork assembly **124** is connected to a bottom portion of the pair of front shocks **122**.

[0076] The rear wheel **127** mounted to the frame **110** by a rear suspension assembly **130**. The rear suspension assembly **130** includes a swing arm **132** and a shock absorber **136**. The swing arm **132** is pivotally mounted at a front thereof to the frame **110**. More specifically, the front of the swing arm **132** is received between lower portions of the lower intermediate frame members **118**. The swing arm **132** pivots relative to the lower intermediate frame members **118** about a swing arm pivot axis **133** that extends through the lower intermediate frame members **118**. As best seen in FIG. **6**, the swing arm **132** includes a swing arm housing **134**, in which is disposed an electric motor **160** and a drivetrain **170**. The swing arm **132** includes a housing cover **137** selectively removable from the housing **134**. The swing arm housing **132** and the housing cover **137** enclose the drivetrain **170**. When the housing cover **137** is in place, the drivetrain **170** is bathed in lubricant within the swing arm **132**.

[0077] The rear wheel **127** is rotatably mounted to the rear end of the swing arm **132** which extends on a left side of the rear wheel **127**. The shock absorber **136** is connected between the swing arm **132** and the frame **110**, specifically to the intermediate frame members **116**. It is contemplated that the relative arrangement of the shock absorber **136** and the frame **110** could vary in different embodiments. The electric motor **160** and the drivetrain **170** will be described in more detail below.

[0078] The vehicle **100** has a straddle seat **140** mounted to the frame **110**, specifically to the rear frame structure **120**, and disposed along the longitudinal centerplane **103**. In the illustrated implementation, the straddle seat **140** is intended to accommodate a single adult-sized rider, i.e. the driver. It is however contemplated that the seat **140** could be longer or that a passenger seat portion could be connected to the rear frame structure **120** in order to accommodate a passenger behind the driver. Depending on the particular implementation, it is also contemplated that the seat **140** could be supported by an assembly of frame members or tubes, a molded portion integrally connected to the seat **140**, or body panels of the motorcycle **100**.

[0079] The vehicle **100** further includes a plurality of body panels **142** for forming the body of the vehicle **100**, illustrated in FIGS. **1** to **3**. The body panels **142** are connected to and supported by the frame members **114**, **116**. The body panels **142** enclose and protect some internal components of the vehicle **100** such as a powerpack **200** (described further below). The vehicle **100** also includes a front fender **144** disposed at the front of the vehicle **100** and extending partially over the front wheel **121**. Rearward of the seat **140**, the vehicle **100** also has rear fender panels **146** extending at least partially over rear wheel **127**. The vehicle **100** includes front headlights **145** attached to the front fork assembly **124** and electrically connected to a battery pack **210** (described further below). The vehicle **100** also has rear braking and indicator lights **147** supported by the rear panels **146** and electrically connected to the battery pack **210**.

[0080] Depending on the particular embodiment, especially between different motorcycle types (trail-style motorcycle or cruiser-style motorcycle for example), the body panels **142** and the fenders **144**, **146** could be different in shape and number. For example, some embodiments of the vehicle **100** could include a mud flab connected to a rear edge of one of the body panels **142**. It is further contemplated that one or both of the fenders **144** and rear panels **146** could be omitted in some cases.

[0081] A driver footrest **126** is disposed on either side of the vehicle **100** and vertically lower than the straddle seat **140** to support the driver's feet. The driver footrests **126** are connected to the frame members **118**. It is contemplated that the footrests **126** could be implemented in various forms other than those illustrated, including but not limited to pegs and footboards. It is contemplated that the vehicle **100** could also be provided with one or more passenger footrests disposed rearward of the driver footrest **126** on each side of the vehicle **100**, for supporting a passenger's feet when a passenger seat portion for accommodating a passenger is connected to the vehicle **100**. A brake pedal **128** is connected to the right driver footrest **126** for braking the vehicle **100**. The brake pedal **128** extends upwardly and forwardly from the right driver footrest **126** such that the driver can actuate the brake pedal **128** with a front portion of the right foot while a rear portion of the right foot remains on the right driver footrest **126**.

[0082] With reference to FIGS. **3** to **5**, each of the front wheel **121** and the rear wheel **127** is provided with a brake assembly **90**. The brake assemblies **90** of the wheels **121**, **127**, along with the brake pedal **128**, form part of a brake system **92**. Each brake assembly **90** is a disc-type brake mounted onto the spindle of the respective wheel **121** or **127**. Other types of brakes are contemplated. Each brake assembly **90** includes a rotor **94** mounted onto the wheel hub and a stationary caliper **96** straddling the rotor **94**. The brake pads (not shown) are mounted to the caliper **96** so as to be disposed between the rotor **94** and the caliper **96** on either side of the rotor **94**. The brake pedal **128**, as well as a hand-operated brake lever **155** described below, are operatively connected to the brake assemblies **90** provided on each of the front wheel **121** and the rear wheel **127**. The brake system **92** further includes a regenerative braking system (not shown) that uses the electric motor **160** as a generator to charge battery cells of the battery pack **210** while slowing the vehicle **100**.

[0083] Returning to FIGS. **1** to **5**, the vehicle **100** includes a handlebar assembly **152** operatively connected to the front fork assembly **124** and disposed in front of the seat **140**. The handlebar assembly **152** is used by the rider to turn the front wheel **121**, via the front fork assembly **124**, to steer the vehicle **100**. Specifically, the handlebar assembly **152** is connected to a top end of the triple clamp assembly **125**. The handlebar assembly **152** and the triple clamp assembly **125** define a steering axis about which the front wheel **121** turns to steer the vehicle **100**. A twist-grip throttle **153** is operatively connected on the right side of the handlebar assembly **152** for controlling vehicle speed. It is contemplated that the twist-grip throttle **153** could be replaced by a throttle lever or some other type of throttle input device. The twist-grip throttle **153** could be disposed on the left side of the handlebar assembly **152** in some embodiments. The handlebar assembly **152** also includes the brake lever **155** on a right side for activating the brake assemblies **90**.

[0084] It is contemplated that the vehicle **100** could include a variety of different features excluded from discussion here, including but not limited to: a windscreen, radio and/or navigational systems, and luggage rack systems.

[0085] The vehicle **100** further includes an electronic powerpack **200**, an electric motor **160**, and a drivetrain **170** for driving the vehicle **100**, specifically the rear wheel **127**. The powerpack **200** will be described in more detail below.

[0086] With reference to FIGS. **6** to **10**, the electric motor **160** is disposed in the swing arm **132**. As the swing arm **132** pivots relative to the frame **110**, the motor **160** moves with the swing arm **132**. In the present embodiment, the motor **160** is a three-phase electric motor **160**. It is contemplated that different types of motors could be used in some embodiments.

[0087] The swing arm housing **134** defines a motor cavity **135** therein in which the motor **160** is disposed (FIG. **8**). The motor **160** is operatively connected to the drivetrain **170** disposed in the swing arm **132**, as is illustrated in FIG. **6**, in which the housing cover **137** has been removed to show the drivetrain **170**. An output shaft **161** of the motor **160** has a gear **171** disposed thereon for engaging the drivetrain **170**. The drivetrain **170** includes a gear wheel **172**, a front sprocket **174** connected to the gear wheel **172** and a rear sprocket **176**. The sprocket **174** engages a belt **178** which in turn engages the sprocket **176**. A belt tensioner **177** presses against the lower part of the belt **178** between the sprockets **174**, **176**.

[0088] Power is provided to the motor **160** by the electronic powerpack **200**. Illustrated in additional detail in FIGS. **6**, **7**, **10**, and **12**, the powerpack **200** is supported by the frame **110**. In the present embodiment, the powerpack **200** is connected to the frame members **118** but different embodiments could have different structural arrangements for connecting the powerpack **200** to the frame **110**.

[0089] The powerpack **200** includes a battery pack **210**. The battery pack **210** includes a battery housing **220**. The battery housing **220** is fastened to the frame **110** to support the powerpack **200** in the illustrated embodiment. As best seen in FIG. **13**, in the present embodiment, the battery pack **210** is completely forward of the swing arm pivot axis **133**. The battery pack **210** includes a plurality of battery cells **230** housed in the battery housing **220**, illustrated schematically in FIGS. **6** and **16**. Depending on the particular implementational details of a given embodiment of the vehicle **100**, the specific implementation details of the battery pack **210** and/or the plurality of battery cells **230** could vary. For example, battery cells could vary in nominal energy capacity, usable energy capacity, discharge rate, cell chemistry and cell type.

[0090] The powerpack **200** includes a charger **250** connected to the battery pack **210**. The charger **250** includes a charger housing **252** surrounding internal electronic components (not shown) of the charger **250**. The charger **250** is mounted to the battery housing **220**. Specifically, the charger housing **252** is fastened to the battery housing **220** and is disposed on a top side of the battery housing **220**. It is contemplated that the location of the charger **250** relative to the battery pack **210** could vary.

[0091] The charger **250** is electrically connected to the battery cells **230** for supplying charge to the battery cells **230**. The vehicle **100** includes a socket **258** electrically connected to the charger **250** for electrically connecting to an external power source for providing electricity to the charger **250** for charging the battery cells **230**. The socket **258** is disposed generally rearward of the charger **250** and extends at least partially through one of the body panels **142**, but the specific location could vary.

[0092] The powerpack **200** also includes an inverter **260** disposed on a left side of the battery pack **210**. The inverter **260** includes an inverter housing **262** which is fastened to the battery housing **220**, specifically along a left side of the battery housing **220**. As such, the inverter **260** is mounted to the battery housing **220**. In some embodiments, it is contemplated that the inverter **260** could be disposed on a different side of the battery pack **210**.

[0093] In order to electrically connect to the battery cells **230** in the battery pack **210**, the inverter

260 includes an electric connector **261** disposed on an exterior of the invertor housing **262** (see FIG. **18**). The battery pack **210** includes an electric connector **215** electrically connected to the battery cells **230** and disposed on an exterior of the battery housing **220**, specifically on a left side of the housing **220** (see FIG. **17**). When the vehicle **100** is in operation, the inverter **260** receives electric power from the battery cells **230** via the electric connector **215** and the electric connector **261**.

[0094] The connector **215** is arranged to receive the connector **261** of the inverter **260**, such that the electric connector **215** and the electric connector **261** are selectively connected together for managing electricity flow from the battery pack **210** to other electronic components of the vehicle **100**. As can be seen in at least FIG. **6**, the inverter **260** is electrically connected to the three-phase motor **160** via three power cords **165** connected to three outlets **269** of the inverter **260**. The number of cords or type of electrical connection between the inverter **260** and the motor **160** could vary in different embodiments. While the inverter **260** connects directly to the battery pack **210** in the present embodiment, it is contemplated that the inverter **260** could be separated and spaced from the battery pack **210** and electrically connected to the battery cells **230** via power cords or the like.

[0095] According to non-limiting embodiments of the present technology, the vehicle **100** includes a cooling circuit **290**, illustrated schematically in FIG. **11**, for cooling electronic components of the vehicle, including the powerpack **200** and the motor **160**. With additional reference to FIGS. **12** to **18**, the cooling circuit **290** is in the form of a closed fluid cooling loop **290** for absorbing heat from the motor **160** and components of the powerpack **200**. Heat transfer in the cooling circuit **290** is provided by a liquid coolant, generally a glycol-water coolant, although it is contemplated that different liquid coolants could be utilized. It is noted that while liquid coolant is provided, some gases may also be present in the cooling circuit **290**, due to phase transitions or air infiltrations.

[0096] In some embodiments, it is contemplated that the cooling circuit **290** could be limited to the powerpack **200** and cooling of the motor **160** could be provided by other means. It is also contemplated that some components of the powerpack **200** could be omitted from the cooling circuit **290** and cooling could be provided by other means. For example, some components of the vehicle **100** could be cooled through air cooling.

[0097] The vehicle **100** includes a coolant reservoir **270** connected to the powerpack **200** and fluidly connected to the cooing circuit **290**. The reservoir **270** receives liquid coolant therein and supplies coolant to the cooling circuit **290**. It is noted that the cooling circuit **290** is considered a “closed loop” in that the coolant flowing through the cooling circuit **290** absorbs heat from heat-generating components and radiates that heat away using heat exchangers (described below) without exchanging the coolant fluids. The reservoir **270** includes a reservoir cap **272** selectively connected thereto. The reservoir **270** provides for coolant to be refilled or supplemented if necessary. When the cap **272** is removed, additional coolant fluid can be added to the reservoir **270** to supplement the fluid level of coolant in the cooling circuit **290**. It is contemplated that the coolant reservoir **270** could be omitted in some embodiments and that coolant fluid could be added elsewhere in the cooling circuit **290**.

[0098] As can be seen in FIG. **12**, the reservoir **270** is disposed partially forward of the powerpack **200**. It is contemplated that the exact positioning of the reservoir **270** could vary in different embodiments. The reservoir **270** is supported by the powerpack **200**; specifically the reservoir **270** is connected to the charger **250** (described further below). It is contemplated that the reservoir **270** could be supported by a different component of the powerpack **200** or the vehicle **100**. For example, it is contemplated that the reservoir **270** could be connected to and/or supported by the frame **110** in some embodiments.

[0099] With reference to FIGS. **13** to **15**, the charger housing **252** defines a charger cooling channel **254** therein. The charger cooling channel **254** extends generally horizontally along a bottom side of the charger **250**. As such, the charger cooling channel **254** extends along a side of the charger **250** facing an interior of the powerpack **200**. The shape of the present embodiment of the channel **254** is illustrated cross-section in FIG. **15**, but it is contemplated that the exact shape could vary.

[0100] As seen in FIG. **13**, the charger **250** includes a channel inlet **253** that is defined in a front, bottom portion of the housing **252**. The inlet **253** is fluidly connected to the channel **254**. The inlet **253** extends downward and forward, into the coolant reservoir **270**, thereby forming a rigid fluid connection between the charger cooling channel **254** and the coolant reservoir **270**. In such an arrangement, coolant flows from the reservoir **270** to the charger cooling channel **254** without the use of piping or hose between the reservoir **270** and the charger **250**. A channel outlet **255** (FIGS. **14** and **15**) is defined on the bottom side of the charger housing **252** and is fluidly connected to the channel **254**.

[0101] With reference to FIGS. **16** and **18**, the inverter housing **262** defines an inverter cooling channel **264** therein. The inverter cooling channel **264** extends generally vertically along and across a right side of the inverter **260**. As such, the inverter cooling channel **264** extends along a side of the inverter **260** facing the interior of the powerpack **200**. The shape of the present embodiment of the channel **264** is illustrated in FIGS. **16** and **18**, but it is contemplated that the exact shape could vary.

[0102] The inverter **260** defines a channel inlet **265** on a top side of the housing **262**, fluidly communicating with the channel **264**. As can be seen in FIG. **16**, the inlet **265** is disposed below the outlet **255** of the charger cooling channel **254**. In the present embodiment, the powerpack **200** further includes a rigid tube **259**, also referred to as a ring **259**, for fluidly and rigidly connecting the charger cooling channel **254** to the inverter cooling channel **264**. It is contemplated that the housings **252**, **262** could be formed to fit sealingly directly together, omitting the tube **259**. The inverter **260** also defines a channel outlet **267** on a right side of the housing **262**, fluidly communicating with the channel **264**.

[0103] As is illustrated in more detail in FIGS. **13** and **16**, the battery housing **220** defines a battery cooling channel **226** therein. The battery cooling channel **226** extends through a center portion of the battery pack **210**, in the interior of the powerpack **200**. As can be seen in the cross-sectional views of FIGS. **13** and **16**, the battery cooling channel **226** includes a plurality of fins extending inward from the housing **220** and coolant fluid flows along a longitudinal direction through a center portion **227** of the housing **220**, along the direction of the centerplane **103**, as well as along a vertical/lateral plane of the vehicle **100** (orthogonal to the centerplane **103**), descending toward a battery cooling channel outlet **217** (FIG. **17**). By being disposed in the center portion **227** of the housing **220**, the channel **226** is in thermal communication with banks of battery cells **230** disposed on both a right side of the channel **226** and a left side of the channel **226**.

[0104] The battery pack **210** includes a channel inlet **223** formed by the battery housing **220**, the inlet **223** fluidly communicating with the channel **226** (see FIGS. **16** and **17**). The housing **220** also defines a channel portion **224** extending from the inlet **223** to the generally vertically extending cooling channel **226**. In some embodiments, the channel portion **224** could be omitted, with the inlet **223** aligning with a top portion of the vertically extending channel **226** for example.

[0105] As can be seen in FIG. **16**, the outlet **267** of the inverter cooling channel **264** is connected to the inlet **223** of the battery cooling channel **226**. The housings **262**, **220** fit sealingly together, such that the inverter cooling channel **264** and the battery cooling channel **226** fluidly connected together by a rigid fluid connection formed by the housings **262**, **220**. In the illustrated embodiment, the electric connector **261** and the inverter channel outlet **267** are disposed on a same side of the inverter **260**. In this arrangement, both the cooling connection and the electrical connection between the inverter **260** and the battery pack **210** are internal to the powerpack **200**. As is further illustrated in FIG. **16**, connections between the battery cooling channel **226**, the inverter cooling channel **264**, and the charger cooling channel **254** are internal to the powerpack **200**. In this way, no hoses are required to make these connections, and the connections between the channels **226**, **264**, **254** are generally protected from interference from external to the vehicle **100**.

[0106] It is noted that the terms “inlet” and “outlet” are not meant to limit the direction of flow through the charger **250**, the inverter **260**, and the battery pack **210**. In embodiments where the direction of flow of coolant through the cooling circuit **290** is reversed, coolant would flow into the openings labelled outlets, including the charger channel outlet **255**, the inverter channel outlet **267**, and the battery channel outlet. Similarly, with the flow through the cooling circuit **290** reversed, coolant could flow out of the openings labelled inlets, including the charger channel inlet **253**, the inverter channel inlet **265**, and the battery channel inlet **223**.

[0107] With reference to FIGS. **8** and **12**, the electric motor **160** includes a motor housing **162**. The motor housing **162** includes a channeled outer surface **164** in thermal contact with at least some internal components, including at least a stator and a rotor (not shown), of the motor **160**. The vehicle **100** thus includes a motor cooling channel **168** (shown schematically in FIG. **11**) for cooling the electric motor **160**. The channel **168** is formed between the channeled outer surface **164** and an internal surface of the motor cavity **135** in which the motor **160** is disposed. The motor cooling channel **168** formed by the channeled outer surface **164** forms a spiral winding around an exterior of the electric motor **160**. In some embodiments, the shape of the channel **168** could vary.

[0108] The motor cooling channel **168** is fluidly connected to the battery cooling channel **226** and forms a portion of the cooling circuit **290**. Specifically, the motor cooling channel **168** is connected to the channel outlet **217** of the battery cooling channel **226** by a flexible tube **167**, also referred to as a hose **167**. With reference to FIGS. **10**, **12**, **13** and **15**, with the vehicle **100** at rest, a front end of the hose **167** is forward and vertically higher than the swing arm pivot axis **133** and a rear end of the hose **167** is rearward of the swing arm pivot axis **133** and in part vertically lower than the swing arm pivot axis **133**. The hose **167** extends in a space defined laterally between the frame members **118** and longitudinally between the battery pack **210** and the motor **160**. The swing arm pivot axis **133** extends through this space. As the motor **160** moves with the swing arm **132** relative to the frame **110**, and thus the powerpack **200**, flexible connections are needed to maintain fluid connection between portions of the cooling circuit **290** in the powerpack **200** and portions of the cooling circuit **290** in thermal communication with the motor **160**. To reduce the movement of the hose **167** as the motor **160** moves with the swing arm **132** relative to the frame **110**, part of the hose **167** is disposed at a radial distance from the swing arm pivot axis **133** that is less than twice a diameter of the hose **167**. In other embodiments, this radial distance is less than **1**.**5** times the diameter of the hose **167**. In other embodiments, this radial distance is less than the diameter of the hose **167**.

[0109] With reference to FIGS. **9**, **12**, and **13**, the vehicle **100** also includes a coolant pump **278** for circulating cooling through and forming a portion of the coolant circuit **290**. In the illustrated embodiment, the coolant pump **278** is electrically connected to the battery pack **210** for powering the pump **278**. The pump **278** is disposed partially rearward of the powerpack **200** in the present embodiment, although it is contemplated that placement of the pump **278** could vary. The pump **278** is fluidly connected to the motor cooling channel **168** by a flexible tube **169**, also referred to as a hose **169**. With reference to FIGS. **10**, **12**, **13** and **15**, with the vehicle **100** at rest, a front end of the hose **169** is forward and vertically higher than the swing arm pivot axis **133** and a rear end of the hose **169** is rearward of the swing arm pivot axis **133** and in part vertically lower than the swing arm pivot axis **133**. The hose **169** also extends in the space defined laterally between the frame members **118** and longitudinally between the battery pack **210** and the motor **160**. As is noted above, the motor **160** pivots with the swing arm **132** relative to the frame **110** and further relative to the pump **278** which is also supported by the frame **110**. The connection between the motor **160** and the pump **278** is thus also required to be flexible. To reduce the movement of the hose **169** as the motor **160** moves with the swing arm **132** relative to the frame **110**, part of the hose **169** is disposed at a radial distance from the swing arm pivot axis **133** that is less than twice a diameter of the hose **169**. In other embodiments, this radial distance is less than **1**.**5** times the diameter of the hose **169**. In other embodiments, this radial distance is less than the diameter of the hose **169**.

[0110] The vehicle **100** further includes two radiators for cooling the coolant fluid: a left radiator **280** disposed on a left side of the vehicle **100** and a right radiator **282** disposed on a right side of the vehicle **100**. In the illustrated embodiment, the radiators **280**, **282** are disposed partially forward of the powerpack **200**, although exact placement could vary in different embodiments. Each radiator **280**, **282** is fluidly connected to the cooling channels **254**, **264**, **226** of the powerpack **200** and forms a portion of the cooling circuit **290**. Each radiator **280**, **282** is arranged to receive airflow thereover during operation of the vehicle **100** to radiate heat away from the coolant via the radiators **280**, **282**.
[0111] The vehicle **100** includes two flexible tubes **285**, with each tube **285** being connected between a corresponding one of the radiators **280**, **282** and the pump **278**. Each tube **285** extends along the corresponding right and left side of the powerpack **200**. The left tube **285** is disposed generally vertically below the inverter **260**, although exact placement of the tubes **285** could vary. The tubes **285**, also referred to as hoses **285**, further form a portion of the cooling circuit **290**, as is illustrated schematically in FIG. **11**. In at least some embodiments, the tubes **285** could be implemented as rigid tubing elements.
[0112] In the illustrated embodiment, the right radiator **282** includes a fan **284** connected to the housing of the radiator **282** to aid in increasing the cooling efficiency of the radiator **282**. Depending on the embodiment, it is contemplated that the left radiator **280** could additionally or alternatively include a fan connected thereto. It is also contemplated that the fan **284** could be omitted in some cases.
[0113] With reference to at least FIG. **12**, the vehicle **100** also includes flexible tubing components **287**, also referred to as tubes **287** or hoses **287**, fluidly connecting the radiators **280**, **282** to the coolant reservoir **270** and forming a portion of the cooling circuit **290**. As such, the tubes **287** fluidly connect each radiator **280**, **282** to the battery cooling channel **226** via the coolant reservoir **270**.
[0114] Returning to FIG. **11**, flow of the cooling circuit **290** is schematically illustrated. Beginning at the coolant reservoir **270** (for simplicity of description), coolant flows from the reservoir **270** into the charger cooling channel **254**. Coolant then flows into the inverter cooling channel **264** and then subsequently into the battery cooling channel **226**. As is noted before, the coolant reservoir **270**, the charger cooling channel **254**, the inverter cooling channel **264**, and the battery cooling channel **226** are rigidly fluidly connected together, without the use of tubes or hoses. From the battery cooling channel **226**, coolant flows through the flexible tube **167** to the motor cooling channel **168**. Coolant then flows from the channel **168** through the flexible tube **169** to the pump **278**. Coolant is then subsequently returned by the pump **278** to forward portions of the vehicle **100** via tubes **285** on each of the right and left sides of the vehicle **100** to the radiators **280**, **282**. Having been at least partially cooled, coolant is then returned to the reservoir **270** via the hoses **287**.
[0115] As is noted above, the direction of coolant flow could be reversed in at least some embodiments. The order of some components forming the coolant circuit **290** could be changed in some embodiments. As one non-limiting example, it is contemplated that coolant could flow through the radiators **280**, **282** before the pump **278**. While the order of the components along the cooling circuit **290** can vary, in the present embodiment the radiators **280**, **282** are upstream from the cooling channels **254**, **264**, **226** of the powerpack **200**. In this way, the powerpack components **250**, **260**, **210** which require more cooling may exchange more heat with the coolant in the cooling circuit **290** than subsequent components such as the motor **160** which are less sensitive to heating.
[0116] With reference to FIG. **19**, another non-limiting embodiment of a vehicle utilizes the powerpack **200** described above, specifically an electric snowmobile **300**. The snowmobile **300** includes a forward end **312** and a rearward end **314** that are defined consistently with a forward travel direction of the snowmobile **300**. The snowmobile **300** includes a frame **316** that includes a tunnel **318**, a motor cradle portion **320** and a front suspension assembly portion **322**.
[0117] The snowmobile **300** includes the powerpack **200** (shown schematically) and an electric

motor **324** (shown schematically), carried by the motor cradle portion **320** of the frame **316**. Two skis **326** (a right ski **326** being illustrated) are positioned at the forward end **312** of the snowmobile **300** and are attached to the front suspension assembly portion **322** of the frame **316** through front suspension assemblies **328**. A steering device in the form of handlebar **336** is attached to the upper end of the steering column **334** to allow a driver to rotate the skis **326**, in order to steer the snowmobile **300**.

[0118] An endless drive track **338** is disposed generally under the tunnel **318** and is operatively connected to the motor **324**. The endless drive track **338** is driven to run about a rear suspension assembly **342** for propulsion of the snowmobile **300**. The rear suspension assembly **342** includes a pair of slide rails **344** in sliding contact with the endless drive track **338**. The rear suspension assembly **342** also includes a plurality of shock absorbers **346**. Suspension arms **348** and **350** are provided to attach the slide rails **344** to the frame **316**. A plurality of idler wheels **352** are also provided in the rear suspension assembly **342**. Other types and geometries of rear suspension assemblies are also contemplated.

[0119] At the forward end **312** of the snowmobile **300**, fairings **354** enclose the powerpack **200**. The fairings **354** include a hood and one or more side panels that can be opened to allow access to the powerpack **200** and/or the motor **324** when this is required, for example, for inspection or maintenance of the powerpack **200** and/or the motor **324**. A windshield **356** is connected to the fairings **354** near the forward end **312** of the snowmobile **300**. Alternatively, the windshield **356** could be connected directly to the handlebar **336**. The windshield **356** acts as a wind screen to lessen the force of the air on the driver while the snowmobile **300** is moving forward.

[0120] A straddle seat **358** is positioned over the tunnel **318**. Two footrests **360** are positioned on opposite sides of the snowmobile **300** below the seat **358** to accommodate the driver's feet. A snow flap **319** is disposed at the rear end **314** of the snowmobile **300**. The tunnel **318** consists of one or more pieces of sheet metal arranged to form an inverted U-shape that is connected at the front to the motor cradle portion **320** and extends rearward therefrom.

[0121] The snowmobile **300** has other features and components which would be readily recognized by one of ordinary skill in the art, further explanation and description of these components will not be provided herein.

[0122] The vehicles **100**, **300** implemented in accordance with some non-limiting implementations of the present technology can be represented as follows, presented in numbered clauses.

[0123] CLAUSE **1**: An electric vehicle (**100**, **300**) comprising: [0124] a frame (**110**); [0125] at least one ground-engaging member (**121**, **127**) operatively connected to the frame (**110**); [0126] an electric motor (**160**) operatively connected to the at least one ground-engaging member (**121**, **127**); [0127] an electric powerpack (**200**) supported by the frame (**110**), the powerpack (**200**) including: [0128] a battery pack (**210**) including: [0129] a battery housing (**220**) defining a battery cooling channel (**226**); and [0130] a plurality of battery cells (**230**) housed in the battery housing (**220**); [0131] a charger (**250**) electrically connected to the plurality of battery cells (**230**), the charger (**250**) including: [0132] a charger housing (**252**) defining a charger cooling channel (**254**); [0133] an inverter (**260**) electrically connected to the plurality of battery cells (**230**), the inverter (**260**) including: [0134] an inverter housing (**262**) defining an inverter cooling channel (**264**), [0135] the battery cooling channel (**226**), the charger cooling channel (**254**), and the inverter cooling channel (**264**) being fluidly connected together by rigid fluid connections; [0136] at least one radiator (**280**, **282**) fluidly connected to the battery cooling channel (**226**); [0137] a cooling circuit (**290**) being formed at least in part by the battery cooling channel (**226**), the charger cooling channel (**254**), the inverter cooling channel (**264**), and the at least one radiator (**280**, **282**); and [0138] a pump (**278**) fluidly connected to the cooling circuit (**290**) for pumping liquid coolant through the cooling circuit (**290**).

[0139] CLAUSE 2. The vehicle (**100**, **300**) of clause 1, further comprising a motor cooling channel (**168**) for cooling the electric motor (**160**) fluidly connected to the battery cooling channel (**226**)

and forming a portion of the cooling circuit (**290**).
[0140] CLAUSE 3. The vehicle (**100**, **300**) of clause 2, further comprising: a swing arm (**132**) pivotally connected to the frame (**110**), the swing arm (**132**) including a swing arm housing (**134**); and [0141] wherein: [0142] the electric motor (**160**) includes a motor housing (**162**); [0143] the motor housing (**162**) includes a channeled outer surface (**164**) in thermal contact with at least some internal components of the electric motor (**160**); [0144] the swing arm housing (**134**) defines a motor cavity (**135**); [0145] the electric motor (**160**) being disposed in the motor cavity (**135**); and [0146] the motor cooling channel (**168**) is formed between the channeled outer surface (**164**) of the motor housing (**162**) and an internal surface of the motor cavity (**135**).
[0147] CLAUSE 4. The vehicle (**100**, **300**) of clause 3, further comprising at least one flexible tube (**167**, **169**) fluidly connecting the motor cooling channel (**168**) to the pump (**278**).
[0148] CLAUSE 5. The vehicle (**100**, **300**) of clause 3, wherein a channel formed by the channeled outer surface (**164**) forms a spiral winding around an exterior of the electric motor (**160**).
[0149] CLAUSE 6. The vehicle (**100**, **300**) of any one of clauses 1 to 5, wherein: [0150] the inverter housing (**262**) is fastened to the battery housing (**220**); and [0151] at least one of an inlet and an outlet of the inverter cooling channel (**264**) is connected to at least one of an inlet and an outlet of the battery cooling channel (**226**).
[0152] CLAUSE 7. The vehicle (**100**, **300**) of clause 6, wherein: [0153] the battery pack (**210**) includes a first electric connector (**215**) electrically connected to the plurality of battery cells (**230**) and disposed on an exterior of the battery housing (**220**); [0154] the inverter (**260**) includes a second electric connector (**261**) disposed on an exterior of the invertor housing (**162**); [0155] the first electric connector (**215**) and the second electric connector (**261**) are selectively connected together; and [0156] when the vehicle (**100**, **300**) is in operation, the inverter (**260**) receives electric power from the plurality of battery cells (**230**) via the first electric connector (**215**) and the second electric connector (**261**).
[0157] CLAUSE 8. The vehicle (**100**, **300**) of clause 7, wherein the second electric connector (**261**) and the at least one of the inlet and the outlet of the inverter cooling channel (**264**) are disposed on a same side of the inverter (**260**).
[0158] CLAUSE 9. The vehicle (**100**, **300**) of any one of clauses 1 to 8, wherein the charger (**250**) and the inverter (**260**) are mounted to the battery housing (**220**).
[0159] CLAUSE 10. The vehicle (**100**, **300**) of clause 9, further comprising a coolant reservoir (**270**) connected to the powerpack (**200**).
[0160] CLAUSE 11. The vehicle (**100**, **300**) of any one of clauses 1 to 10, wherein at least one of the rigid fluid connections connecting the battery cooling channel (**226**), the charger cooling channel (**254**), and the inverter cooling channel (**264**) together is formed by one of the charger housing (**252**), the battery housing (**220**), and the inverter housing (**262**).
[0161] CLAUSE 12. The vehicle (**100**, **300**) of any one of clauses 1 to 11, wherein connections between the battery cooling channel (**226**), the charger cooling channel (**254**), and the inverter cooling channel (**264**) are internal to the electric powerpack (**200**).
[0162] CLAUSE 13. The vehicle (**100**, **300**) of any one of clauses 1 to 12, wherein: [0163] the charger cooling channel (**254**) extends along a side of the charger (**250**) facing an interior of the powerpack (**200**); and [0164] the inverter cooling channel (**264**) extends along a side of the inverter (**260**) facing the interior of the powerpack (**200**).
[0165] CLAUSE 14. The vehicle (**100**, **300**) of clause 13, wherein the battery cooling channel (**226**) extends through a center portion of the battery pack (**210**) in the interior of the powerpack (**200**).
[0166] CLAUSE 15. The vehicle (**100**, **300**) of any one of clauses 1 to 14, wherein the at least one radiator (**280**, **282**) includes: [0167] a left radiator (**280**) disposed on a left side of the vehicle (**100**, **300**); and [0168] a right radiator (**282**) disposed on a right side of the vehicle (**100**, **300**).
[0169] CLAUSE 16. The vehicle (**100**, **300**) of clause 15, further comprising a plurality of flexible

tubing components (**287**) fluidly connecting the left radiator (**280**) and the right radiator (**282**) to the at least one battery cooling channel (**226**) and forming a portion of the cooling circuit (**290**).
[0170] CLAUSE 17. The vehicle (**100**, **300**) of clause 15, wherein at least one of the left radiator (**280**) and the right radiator (**282**) includes a fan (**284**) connected to a radiator housing thereof.
[0171] CLAUSE 18. The vehicle (**100**, **300**) of any one of clauses 1 to 17, further comprising a coolant reservoir (**270**) fluidly connected to the cooling circuit (**290**).
[0172] CLAUSE 19. The vehicle (**100**, **300**) of clause 18, wherein the charger housing (**252**) includes at least one of an inlet and an outlet configured to be sealingly fit into at least one of an outlet and an inlet of the coolant reservoir (**270**).
[0173] CLAUSE 20. The vehicle (**100**, **300**) of clause 18, wherein: [0174] the coolant reservoir (**270**) is disposed at least partially forward of the electric powerpack (**200**); and [0175] the pump (**278**) is disposed at least partially rearward of the electric powerpack (**200**).
[0176] CLAUSE 21. The vehicle (**100**, **300**) of any one of clauses 1 to 20, further comprising: [0177] a motor cooling channel (**168**) for cooling the electric motor (**160**) fluidly connected to the battery cooling channel (**226**), and [0178] a coolant reservoir (**270**) connected to the powerpack (**200**); and [0179] wherein: [0180] the at least one radiator (**280**, **282**) includes a pair of radiators (**280**, **282**), and [0181] the cooling circuit (**290**) is further formed by at least the pair of radiators (**280**, **282**), the motor cooling channel (**168**), and the coolant reservoir (**270**).
[0182] CLAUSE 22. The vehicle (**100**) of any one of clauses 1 to 21, further comprising: [0183] a straddle seat (**140**); and [0184] wherein: [0185] the at least one ground-engaging member (**121**,**127**) includes: [0186] at least one front ground-engaging member (**121**) disposed at least in part forward of the electric powerpack (**200**), and [0187] at least one rear ground-engaging member (**127**) disposed at least in part rearward of the electric powerpack (**200**).
[0188] CLAUSE 23. The vehicle (**100**) of any one of clauses 1 to 22, wherein: [0189] the at least one ground-engaging member (**121**,**127**) includes: [0190] a front wheel (**121**) disposed at least in part forward of the electric powerpack (**200**), and [0191] a rear wheel (**127**) disposed at least in part rearward of the electric powerpack (**200**); and [0192] the vehicle (**100**) is an electric motorcycle (**100**).
[0193] CLAUSE 24. The vehicle (**300**) of any one of clauses 1 to 22, wherein: [0194] the vehicle (**300**) is an electric snowmobile (**300**); and [0195] the at least one ground-engaging member includes two skis (**326**).
[0196] CLAUSE 25. An electric vehicle (**100**) comprising: [0197] a frame (**110**), the frame (**110**) having a pair of frame members (**118**); [0198] a front ground-engaging member (**121**) operatively connected to the frame (**110**); [0199] a swing arm (**132**) pivotally connected to the pair of frame members (**118**) about a swing arm pivot axis (**133**), the swing arm pivot axis (**133**) extending through the pair of frame members (**118**); [0200] a rear ground-engaging member (**127**) operatively connected to the swing arm (**132**); [0201] an electric motor (**160**) mounted to the swing arm (**132**) and operatively connected to the rear ground-engaging member (**127**); [0202] a motor cooling channel (**168**) in thermal communication with the electric motor (**160**) for cooling the electric motor (**160**); [0203] a battery pack (**210**) supported by the frame (**110**); [0204] a battery cooling channel (**226**) in thermal communication with the battery pack (**210**) for cooling the battery pack (**210**); and [0205] at least one hose fluidly (**167**, **169**) connected between the battery cooling channel (**226**) and the motor cooling channel (**168**), the at least one hose (**167**, **169**) extending in a space defined laterally between the frame members (**118**) and longitudinally between the battery pack (**210**) and the motor (**160**).
[0206] CLAUSE 26. The vehicle (**100**) of clause 25, wherein the swing arm pivot axis (**133**) extends through the space defined laterally between the frame members (**118**) and longitudinally between the battery pack (**210**) and the motor (**160**).
[0207] CLAUSE 27. The vehicle (**100**) of clause 25 or 26, wherein, with the vehicle (**100**) at rest: [0208] a front end of the at least one hose (**167**, **169**) is forward of the swing arm pivot axis (**133**);

[0209] the front end of the at least one hose (**167**, **169**) is vertically higher than the swing arm pivot axis (**133**); [0210] a rear end of the at least one hose (**167**, **169**) is rearward of the swing arm pivot axis (**133**); and [0211] the rear end of the at least one hose (**167**, **169**) is at least in part vertically lower than the swing arm pivot axis (**133**).
[0212] CLAUSE 28. The vehicle (**100**) of any one of clauses 25 to 27, wherein part of the at least one hose (**167**, **169**) is disposed at a radial distance from the swing arm pivot axis (**133**) that is less than twice a diameter of the at least one hose (**167**, **169**).
[0213] CLAUSE 29. The vehicle (**100**) of clause 28, wherein the radial distance is less than 1.5 times the diameter of the at least one hose (**167**, **169**).
[0214] CLAUSE 30. The vehicle (**100**) of clause 29, wherein the radial distance is less than the diameter of the at least one hose (**167**, **169**).
[0215] CLAUSE 31. The vehicle (**100**) of any one of clauses 25 to 30, wherein: [0216] the at least one hose (**167**, **169**) includes a first hose (**169**) and a second hose (**167**); [0217] one of the first and second hoses (**167**, **169**) supplying liquid coolant from the battery cooling channel (**226**) to the motor cooling channel (**168**); and [0218] another one of the first and second hoses (**167**, **169**) supplying liquid coolant from the motor cooling channel (**168**) to the battery cooling channel (**226**).
[0219] CLAUSE 32. The vehicle (**100**) of clause 31, further comprising a pump (**278**); and [0220] wherein the pump (**278**) is fluidly connected between the first hose (**169**) and the battery cooling channel (**226**).
[0221] CLAUSE 33. The vehicle (**100**) of any one of clauses 25 to 32, wherein the battery pack (**210**) includes: [0222] a battery housing (**220**) defining the battery cooling channel (**226**); and [0223] a plurality of battery cells (**230**) housed in the battery housing (**220**).
[0224] CLAUSE 34. The vehicle (**100**) of any one of clauses 25 to 33, wherein the battery pack (**210**) is completely forward of the swing arm pivot axis (**133**).
[0225] Modifications and improvements to the above-described implementations of the present technology may become apparent to those skilled in the art. The foregoing description is intended to be exemplary rather than limiting. The scope of the present technology is therefore intended to be limited solely by the scope of the appended claims.

Claims

1. An electric vehicle comprising: a frame; at least one ground-engaging member operatively connected to the frame; an electric motor operatively connected to the at least one ground-engaging member; an electric powerpack supported by the frame, the powerpack including: a battery pack including: a battery housing defining a battery cooling channel; and a plurality of battery cells housed in the battery housing; a charger electrically connected to the plurality of battery cells, the charger including: a charger housing defining a charger cooling channel; an inverter electrically connected to the plurality of battery cells, the inverter including: an inverter housing defining an inverter cooling channel, the battery cooling channel, the charger cooling channel, and the inverter cooling channel being fluidly connected together by rigid fluid connections; at least one radiator fluidly connected to the battery cooling channel; a cooling circuit being formed at least in part by the battery cooling channel, the charger cooling channel, the inverter cooling channel, and the at least one radiator; and a pump fluidly connected to the cooling circuit for pumping liquid coolant through the cooling circuit.
2. The vehicle of claim 1, further comprising a motor cooling channel for cooling the electric motor fluidly connected to the battery cooling channel and forming a portion of the cooling circuit.
3. The vehicle of claim 2, further comprising: a swing arm pivotally connected to the frame, the swing arm including a swing arm housing; and wherein: the electric motor includes a motor housing; the motor housing includes a channeled outer surface in thermal contact with at least some internal components of the electric motor; the swing arm housing defines a motor cavity; the

electric motor being disposed in the motor cavity; and the motor cooling channel is formed between the channeled outer surface of the motor housing and an internal surface of the motor cavity.
4. The vehicle of claim 3, further comprising at least one flexible tube fluidly connecting the motor cooling channel to the pump.
5. The vehicle of claim 3, wherein a channel formed by the channeled outer surface forms a spiral winding around an exterior of the electric motor.
6. The vehicle of claim 1, wherein: the inverter housing is fastened to the battery housing; and at least one of an inlet and an outlet of the inverter cooling channel is connected to at least one of an inlet and an outlet of the battery cooling channel.
7. The vehicle of claim 6, wherein: the battery pack includes a first electric connector electrically connected to the plurality of battery cells and disposed on an exterior of the battery housing; the inverter includes a second electric connector disposed on an exterior of the invertor housing; the first electric connector and the second electric connector are selectively connected together; and when the vehicle is in operation, the inverter receives electric power from the plurality of battery cells via the first electric connector and the second electric connector.
8. (canceled)
9. The vehicle of claim 1, wherein the charger and the inverter are mounted to the battery housing.
10. (canceled)
11. The vehicle of claim 1, wherein at least one of the rigid fluid connections connecting the battery cooling channel, the charger cooling channel, and the inverter cooling channel together is formed by one of the charger housing, the battery housing, and the inverter housing.
12. The vehicle of claim 1, wherein connections between the battery cooling channel, the charger cooling channel, and the inverter cooling channel are internal to the electric powerpack.
13. The vehicle of claim 1, wherein: the charger cooling channel extends along a side of the charger facing an interior of the powerpack; and the inverter cooling channel extends along a side of the inverter facing the interior of the powerpack.
14. The vehicle of claim 13, wherein the battery cooling channel extends through a center portion of the battery pack in the interior of the powerpack.
15. The vehicle of claim 1, wherein the at least one radiator includes: a left radiator disposed on a left side of the vehicle; and a right radiator disposed on a right side of the vehicle.
16. The vehicle of claim 15, further comprising a plurality of flexible tubing components fluidly connecting the left radiator and the right radiator to the at least one battery cooling channel and forming a portion of the cooling circuit.
17. (canceled)
18. The vehicle of claim 1, further comprising a coolant reservoir fluidly connected to the cooling circuit.
19. The vehicle of claim 18, wherein the charger housing includes at least one of an inlet and an outlet configured to be sealingly fit into at least one of an outlet and an inlet of the coolant reservoir.
20. (canceled)
21. The vehicle of claim 1, further comprising: a motor cooling channel for cooling the electric motor fluidly connected to the battery cooling channel, and a coolant reservoir connected to the powerpack; and wherein: the at least one radiator includes a pair of radiators, and the cooling circuit is further formed by at least the pair of radiators, the motor cooling channel, and the coolant reservoir.
22. The vehicle of claim 1, further comprising: a straddle seat; and wherein: the at least one ground-engaging member includes: at least one front ground-engaging member disposed at least in part forward of the electric powerpack, and at least one rear ground-engaging member disposed at least in part rearward of the electric powerpack.

23. The vehicle of claim 1, wherein: the at least one ground-engaging member includes: a front wheel disposed at least in part forward of the electric powerpack, and a rear wheel disposed at least in part rearward of the electric powerpack; and the vehicle is an electric motorcycle.
24. The vehicle of claim 1, wherein: the vehicle is an electric snowmobile; and the at least one ground-engaging member includes two skis.
25.-**34**. (canceled)